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Empowering university teachers in higher education: A generative AI-responsive competency framework

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ABSTRACT

The integration of generative artificial intelligence (GenAI) into higher education necessitates a reconceptualization of teacher competencies, moving beyond technical proficiency to encompass pedagogical strategies for fostering critical, ethical, and developmentally appropriate student-AI collaboration. Existing competency frameworks, however, exhibit notable limitations in equipping university teachers with actionable guidance for designing GenAI-mediated learning experiences that cultivate their students' higher-order thinking and subject knowledge. In response, this paper develops and proposes a GenAI-responsive competency framework for university teachers to supplement existing frameworks and address areas that are not sufficiently covered or elaborated. Developed through a systematic analysis of digital and AI-related competency frameworks, the proposed model is grounded in constructivist learning theory, sociological perspectives, and student-centered pedagogy. Its theoretical and practical robustness was further refined through iterative expert review and consultation. The resulting framework comprises four core dimensions: *GenAI Literacy*, *Curriculum/Learning Design*, *Teaching and Learning*, and *Assessment*. Each dimension is articulated through a dual perspective: teachers' own proficiency and their capacity to foster students' critical engagement with GenAI. Competency progression is structured across three developmental levels: *Basic*, *Intermediate*, and *Advanced*, representing a continuum from technical awareness to guided application, and ultimately to critical and creative integration. The proposed framework supports teachers' ongoing professional growth and enhances their ability to facilitate student autonomy, ethical reasoning, and collaborative engagement with GenAI. It provides a structured yet flexible tool for self-assessment, instructional design, and targeted professional development in higher education, thereby advancing the discourse on effective and responsible human-AI collaboration.

1. Introduction

The pervasive integration of generative artificial intelligence (GenAI) into the higher education ecosystem is not only transforming operational infrastructures but also fundamentally recalibrating its core relational dynamics. As faculty and students constitute the central agents within this ecosystem, the teacher-student relationship, together with the pedagogical practices and assessment regimes that structure it, increasingly demands critical re-examination. Acknowledging this imperative, and in alignment with emerging scholarly discourse (e.g., Hwang et al., 2020; Walker & Larson, 2024), this paper introduces a theorized GenAI-responsive teacher competency framework. The framework is conceived as a developmental tool, designed to equip educators with the sophisticated competencies necessary for navigating this reconfigured terrain. Its primary objective is to enable the design of learning experiences that are pedagogically coherent, ethically grounded, and conducive to a critically engaged form of human-AI collaboration. Ultimately, by providing a structured model for professional growth, this contribution seeks to advance the re-conceptualization of teacher preparedness for effective and responsible practice in higher education increasingly characterized by the pervasive integration of AI technologies.

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1.1. Paradigm shift in higher education

Unlike conventional information and communication technology (ICT), GenAI exhibits advanced cognitive capabilities (Qu et al., 2025). These include adaptive learning, contextual information synthesis, autonomous error correction, and heuristic problem-solving. Such capabilities were once considered exclusive to human expertise (Ghasemi et al., 2022; Hwang, 2003; Popenici & Kerr, 2017; Shneiderman, 2020). Through a rapid ideation, feedback, and iteration process, GenAI reconfigures these human-exclusive traits by augmenting learners' creativity and critical thinking and by prompting them to critically evaluate generated content (Rana et al., 2025). This technological revolution spans interdisciplinary domains, including machine learning, natural language processing, architecture design, and computer vision, each contributing distinct affordances that are reshaping educational practices (Liu et al., 2025; Mittal et al., 2024).

Central to GenAI's disruptive potential is its generative epistemology, the ability to autonomously produce original content, pedagogical solutions, and research insights through emergent algorithmic processes (Dai et al., 2023; Wang et al., 2024). On the one hand, tools such as ChatGPT, QuillBot, Claude AI, and DeepSeek grant learners unprecedented agency to decide when, how, and for what purposes to engage with GenAI (Dai et al., 2023; Mawarsih et al., 2025; Mohammed et al., 2025; Nguyen, 2025; Noroozi et al., 2024). On the other hand, GenAI introduces significant challenges related to academic integrity, algorithmic bias, and the potential erosion of foundational learning processes and cognitive autonomy (Nguyen, 2025; Wang et al., 2024; Yusuf et al., 2024). The shift from teacher-prescribed ICT adoption to learner-directed GenAI engagement represents a fundamental restructuring of educational dynamics (Monzon & Hays, 2025). It necessitates a re-examination of the core competencies required for both teachers and students in the GenAI era.

In higher education, teacher competencies typically emphasize disciplinary expertise, facilitating independent inquiry, and developing critical thinking and epistemic agency within flexible and diverse learning environments (Chan & Tsi, 2023; Chen et al., 2025; Dai et al., 2023). However, the rapid integration of GenAI reveals significant insufficiencies of existing competencies. Most existing frameworks were designed prior to the emergence of GenAI and offer limited guidance on fostering meaningful student-AI interactions or leveraging GenAI's potential as a cognitive partner to promote deeper learning (Kohnke et al., 2023; Ng et al., 2023). As GenAI shifts from being a peripheral instructional aid to becoming a transformative force that redefines educational missions, theories, and practices, there is an urgent need to reconceptualize teacher competencies. University teachers must now be prepared to address novel ethical, pedagogical, and technological challenges and to equip students for a future shaped by human-AI collaboration (Chiu et al., 2023).

1.2. Research gap

The emergence of generative AI has stimulated important scholarly work on AI competency frameworks in education, including contributions such as the AI Literacy Framework (Chiu et al., 2024), AI Teacher Competency Standards (Ng et al., 2023), and the AI-TPACK model (Karataş & Ataç, 2024; Ning et al., 2024). These frameworks provide valuable foundations for understanding the technical knowledge and operational skills needed for AI integration. However, as GenAI continues to transform higher education, several areas merit deeper scholarly attention to better support educators in this evolving landscape.

Current research in higher education would benefit from more comprehensive investigation into how technical AI competencies translate into effective pedagogical practices. While existing frameworks adequately address basic operational proficiency, there remains a need to better understand how educators can facilitate meaningful, discipline-specific student-AI collaboration that promotes critical engagement and deep learning (Kohnke et al., 2023; Ng et al., 2023). Additionally, the developmental trajectory of teacher competencies in the GenAI context requires further elaboration. Most existing frameworks present competencies in a relatively static manner, without fully addressing how they might progress across different proficiency levels or adapt to varied disciplinary contexts and institutional settings (Hoidn & Reusser, 2020). Furthermore, as GenAI enables greater student autonomy in learning processes, research has yet to fully explore how educators can effectively scaffold this autonomy. There is limited guidance on pedagogical strategies that support students' responsible and critical use of GenAI while maintaining academic integrity and fostering epistemic growth (Nguyen, 2025; Wang et al., 2024).

These areas represent significant opportunities for advancing our understanding of how to best prepare higher education teachers for the distinctive challenges and possibilities presented by GenAI technologies.

1.3. Research objectives

Building on the identified limitations of existing AI competency frameworks, this paper aims to develop a GenAI-responsive competency framework tailored for university teachers. The proposed framework moves beyond a narrow focus on technical proficiency, articulating the essential competencies required for designing learning experiences that promote students' 21st-century skills, scaffold self-directed learning, and guide the responsible, ethical, and critically informed application of GenAI technologies (Cha et al., 2024; Ng et al., 2023, 2025). By examining the evolving role of GenAI in higher education and systematically addressing gaps in current AI-related competency models, this research seeks to support teachers in creating developmentally appropriate, ethically grounded, and pedagogically robust learning environments.

2. Literature review

Through a systematic examination of the evolution of GenAI in educational contexts, current student engagement patterns, and existing competency frameworks, this section seeks to establish a theoretical foundation for developing a pedagogy-centered, GenAI-responsive competency framework for higher education teachers.

2.1. The evolving integration of GenAI in higher education

The integration of GenAI in higher education reflects a notable shift in academic and institutional attitudes, progressing through three distinct phases: resistance, cautious acceptance, and proactive innovation (McDonald et al., 2025; Yusuf et al., 2024). Initially, both educators and institutions expressed concerns about the risks posed by GenAI, particularly regarding academic integrity, the accuracy and reliability of AI-generated content, and the potential for students' over-reliance on automated tools (Lambert & Stevens, 2024; Rasul et al., 2023). As a result, many institutions responded by regulating or even banning the use of GenAI in academic work, upholding traditional standards of assessment and knowledge production (Cassidy, 2023; Moorhouse et al., 2023).

As GenAI technologies became more pervasive and their value in supporting teaching and learning became increasingly evident, a gradual shift in attitudes emerged. Teachers have begun to acknowledge that GenAI is now an integral part of the academic environment and that outright prohibition is neither practical nor conducive to pedagogical advancement (Batista et al., 2024; Chiu, 2024; Dai et al., 2024). However, despite this growing acceptance, significant challenges persist. Teachers often grapple with uncertainties regarding effective and ethical GenAI integration strategies, insufficient actionable pedagogical frameworks or institutional guidelines, limited professional development opportunities, and concerns related to equity and access (Batista et al., 2024; Francis et al., 2025; Nguyen, 2025). This transitional period is marked by cautious experimentation, as institutions and individuals explore how GenAI can be meaningfully and responsibly embedded into teaching, learning, and assessment practices (Ansari et al., 2024; Chaudhry et al., 2023; Huang et al., 2025; Lu & Ba, 2025; Pan et al., 2025).

Looking forward, the field is characterized by the proactive and strategic integration of GenAI into educational ecosystems. In this phase, teachers are not merely adapting to GenAI but actively leveraging its capabilities to reimagine pedagogical practices, foster new forms of innovation, personalize learning pathways, and enhance collaboration (Chiu, 2024; Kohnke et al., 2025; Msambwa et al., 2025). This means they are moving beyond the use of GenAI for routine tasks, such as automating content generation or administrative functions, and instead engaging in pedagogical redesign (Moundridou et al., 2024; Zhai, 2024). Empirically, teachers are developing new instructional models, assessment strategies, and feedback mechanisms that integrate GenAI in ways that enhance learning, foster critical engagement, and support more dynamic and learner-centered educational experiences (Ba et al., 2025; Chiu, 2024; Moundridou et al., 2024; Xia et al., 2024).

This evolution from resistance to innovation highlights the transformative potential of GenAI in education. However, this progress will not happen automatically without equipping teachers with the necessary competencies to navigate and shape this transition thoughtfully and ethically.

2.2. A critical review of existing AI-related competency frameworks

This section employs a structured analytical approach to evaluate existing AI-related competency frameworks based on four key criteria aligned with the unique demands of GenAI in higher education (Atchley et al., 2024; Cukurova, 2025; Deng & Zhang, 2023). These criteria include: (1) Pedagogical vs. Technical Focus: the extent to which frameworks address teaching strategies versus technical skills; (2) Developmental Perspective: whether they provide progressive pathways for competency growth; (3) Ethical Integration: how they incorporate ethical reasoning and responsible AI use; and (4) Student-Centered Design: the degree to which they consider student-AI interaction and empowerment.

While several frameworks have been introduced to guide teachers in integrating digital and AI technologies into their practice, their alignment with the outlined criteria remains inconsistent. A critical evaluation reveals considerable variation in how these models conceptualize and operationalize the interplay between teachers, students, and AI systems within higher education settings. Notably, few frameworks provide a sufficiently nuanced or adaptable structure to address the dynamic, multi-layered interactions characteristic of GenAI-mediated learning environments. This gap underscores the need for a more comprehensive and context-sensitive framework that systematically addresses the complex and evolving demands faced by university teachers. The following review critically examines the major contributions and limitations of existing frameworks in light of these challenges.

2.2.1. AI Technological Pedagogical Content Knowledge framework

Prior to the emergence of AI technologies, the Technological Pedagogical Content Knowledge (TPACK) framework primarily addressed the integration of traditional ICT, such as multimedia and online platforms, into teaching practices (Koh, 2019). As a structured model, TPACK emphasized the interplay between Content Knowledge (CK), Pedagogical Knowledge (PK), and Technological Knowledge (TK), with intersecting domains (PCK, TCK, TPK) guiding teachers in aligning pedagogy, subject matter, and technological tools (Chai et al., 2013; Koehler & Mishra, 2005). While this framework provided a systematic approach to technology-enhanced learning, its generalizability to general digital technology tools has become problematic with the emergence of GenAI, which exhibits unique qualities, including autonomy, adaptability, sociability, and opaque decision-making processes (Mishra et al., 2023). These affordances have exposed critical limitations in TPACK, particularly its inability to account for AI's ethical implications (e.g., bias, privacy), the knowledge and skills of facilitating the development of student AI literacy, and the dynamic co-construction of knowledge between humans and GenAI systems (Chiu, 2024; Jin et al., 2025; Ning et al., 2024).

To address these challenges, scholars have sought to adapt TPACK to AI-driven education. Early efforts, such as An et al.'s (2023) integration of TPACK with the Unified Theory of Acceptance and Use of Technology (UTAUT), highlighted the role of user acceptance. Another comprehensive reconceptualization, proposed by Celik (2023) through the Intelligent-TPACK scale, introduces AI-specific knowledge domains: Intelligent-TK (e.g., prompt engineering, model limitations), Intelligent-TPK (e.g., AI-powered adaptive pedagogies), and Intelligent-TCK (e.g., discipline-specific AI applications). This model not only expanded TPACK's technological dimension but also emphasized continuous teacher upskilling to keep pace with AI advancements.

Despite these advancements, AI-TPACK frameworks remain constrained by their residual tool-centric perspective, failing to fully acknowledge GenAI as an agentic participant in education (Lan et al., 2025; Lee et al., 2025). For example, empirical studies have revealed persistent challenges, such as teachers' reliance on incidental learning for AI competencies, which leads to misconceptions and uneven classroom implementation (Velander et al., 2023). Other key unresolved issues include the lack of culturally responsive AI integration strategies and insufficient metrics for assessing AI's impact on higher-order thinking. The proposed AI integration model should incorporate Responsible AI principles—such as algorithmic transparency and equity audits—while redefining the roles of teachers and students in human-AI collaborative learning environments (Shneiderman, 2020).

2.2.2. Digital competency framework for educators

The Digital Competency Framework for Educators (DigCompEdu), developed by the European Commission (Caena & Redecker, 2019; Redecker, 2017), provides a comprehensive structure for articulating the digital competencies required of teachers in the 21st century. As illustrated in Fig. 1, the framework maps teachers' competencies into three domains: professional, pedagogic, and learners' competence. Within these domains, six key competences are delineated: professional engagement (e.g., organizational communication and reflective practice), digital resources (e.g., selecting, creating, and managing digital content), teaching and learning (e.g., designing collaborative and self-regulated learning environments), assessment (e.g., implementing digitally supported assessment strategies), empowering learners (e.g., promoting accessibility and personalization), and facilitating learners' digital skills (e.g., fostering information literacy, communication skills, and responsible use of technology).

The DigCompEdu Framework (Redecker, 2017) establishes a tiered model for teachers' technology integration, progressing from basic tool adoption (A1-A2) to strategic implementation (B1-B2) and ultimately leadership in digital innovation (C1-C2) (Table 1). While effectively mapping competency development across six pedagogical domains, the framework requires expansion to address two critical gaps in the AI era: (1) the evolving human-AI partnership dynamics in teaching praxis, and (2) discipline-specific manifestations of GenAI's pedagogical impact. Its current structure, though valuable for foundational digital skills, is insufficient to capture how autonomous technologies reshape teacher roles and decision-making processes in higher education.

2.2.3. Starkey's Digital Age Learning Matrix

Starkey's (2011) Digital Age Learning Matrix offers a structured and influential framework for understanding how digital technologies can support progressive cognitive development in learning. The matrix outlines six ascending levels of learning activity, from basic "doing" to higher-order "creating and sharing knowledge", each associated with increasing cognitive complexity and more intentional uses of technology (Table 2). Grounded in constructivist pedagogy, it positions digital tools

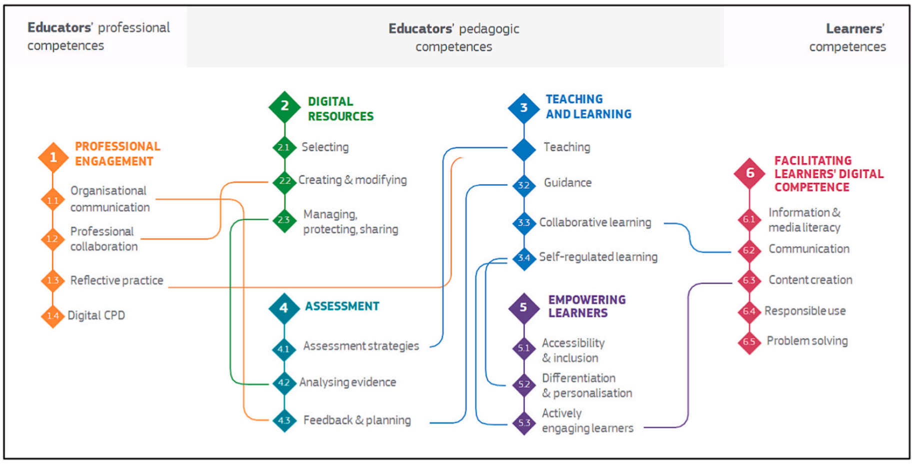

Fig. 1. DigCompEdu Framework (Redecker, 2017, p. 16).

Table 1
DigCompEdu Proficiency Levels (Adapted from Redecker, 2017, p. 30).

Proficiency Level	Role Descriptor	Key Characteristics
A1	Newcomer	Teachers are beginning to explore digital tools. They require guidance to integrate technology into teaching, learning, or assessment.
A2	Explorer	Teachers experiment with basic digital tools in specific contexts but lack a systematic approach.
B1	Integrator	Teachers use digital tools strategically and regularly in teaching/learning. They adapt tools to learners' needs and reflect on their practices.
B2	Expert	Teachers confidently innovate with digital tools, fostering collaboration and creativity. They critically evaluate resources and support peers.
C1	Leader	Teachers lead digital initiatives, mentor others, and shape institutional strategies. They address complex challenges (e.g., accessibility, digital equity).
C2	Pioneer	Teachers drive innovation in digital education, contribute to research, and influence policy. They experiment with emerging technologies (e.g., AI, VR) at scale.

not merely as functional aids but as cognitive mediators that can scaffold critical thinking, creativity, and collaboration when intentionally integrated (Starkey, 2011; Sun et al., 2016). This framework has had a lasting impact on instructional design and assessment by emphasizing alignment between the use of technology and pedagogical goals.

Developed prior to the widespread integration of advanced GenAI, the matrix provides a robust foundation for understanding the progression of technology-mediated learning. However, its original formulation does not fully account for the unique capabilities and roles of AI, such as content generation, conversational interaction, and adaptive feedback, which differ substantially from earlier digital tools. These AI-specific functions introduce new educational dynamics, including opportunities for human–AI knowledge co-construction, as well as new demands such as evaluating AI-generated content, addressing ethical considerations like algorithmic bias, and fostering critical AI literacy.

Therefore, while Starkey's matrix remains highly relevant due to its structured approach to cognitive and technological integration, it presents an opportunity for expansion and evolution in the GenAI era. Further adaptations could incorporate dimensions such as evaluating the authenticity and accuracy of AI-generated outputs, managing cognitive load in human–AI collaboration, and integrating GenAI-specific competencies, all within the unique context of higher education. Such an evolution would preserve the matrix's core educational value while extending its applicability to contemporary GenAI-enhanced learning environments.

2.2.4. UNESCO AI competency framework

The UNESCO AI Competency Framework for Teachers, developed by Cukurova and Miao (2024), represents a significant effort to structure teachers' competencies in navigating the integration of AI into educational contexts. The framework outlines fifteen key skills across five major dimensions: human-centered mindset, ethics of AI, AI foundations and applications, AI pedagogy, and AI for professional development. Each dimension is organized into three progressive proficiency levels: Acquire, Deepen, and Create, depicting a developmental pathway from foundational awareness to active application and innovation (Table 3). For example, within the human-centered mindset dimension, teachers move from promoting human agency to assuming human accountability and ultimately fostering broader social responsibility in the application of AI. Similarly, AI pedagogy evolves from AI-assisted teaching to AI-pedagogy integration, and further to AI-enhanced pedagogical transformation.

A notable strength of the UNESCO framework lies in its clear emphasis on ethical considerations, human rights protection, inclusivity, and sustainability. It encourages teachers not only to understand AI's technical aspects but also to critically engage with ethical controversies, advocate for equity, particularly for marginalized learners, and promote responsible and trustworthy use of AI technologies. Its normative orientation supports strategic policy formulation and national-level teacher development initiatives, aligning AI integration with social values and professional growth objectives.

However, while conceptually robust, the framework tends to remain abstract and policy-driven, offering limited operational guidance for the practical design and implementation of GenAI-enhanced learning environments. It primarily defines "what" competencies teachers should possess but gives less attention to "how" these competencies can be enacted in diverse, dynamic classroom contexts, particularly in fostering students' critical engagement, creative use, and co-construction of knowledge with AI systems. As such, although the framework provides a valuable ethical and developmental foundation, further elaboration is needed to translate its principles into actionable pedagogical practices responsive to the realities of GenAI-powered education (Lan et al., 2025).

Building on concerns about human agency and technological autonomy, Shneiderman (2020, 2022) introduced the Human-Centered AI (HCAI) framework, conceptualizing automation along two dimensions: the degree of human versus computer control and the level of automation, ranging from low to high. Extending this idea, Cukurova (2025) proposed the AIED-HCD model, which reframes AI as a "computational

Table 2
Digital Age Learning Matrix (Starkey, 2011, p. 29).

Aspect of learning	Doing	Thinking about connections	Thinking about concepts	Critiquing and evaluating	Creating knowledge	Sharing knowledge
	Explanation of the levels of learning					
Digital technology use	Isolated information. Focus on completing a measurable task	Connecting thinking. Simple connections made within a context. Compare and share	Develop conceptual understanding of 'big ideas'	Evaluating and critiquing to explore the limitations and the potential of information, sources or process	Creativity-Applying ideas, processes and/or experiences to develop a new reality	Sharing the new knowledge through authentic contexts and gaining feedback to measure value
Accessing information	Accessing: Pictures, Graphs, Movies, and Data Information	Information from more than one source is connected or compared in the analysis	Information explicitly develops conceptual understanding	Information and sources are critiqued and evaluated	New conceptual understanding is developed. Building on or linking accessed information	The value of the product is determined by the quality and quantity of feedback from beyond the classroom environment. Learning occurs when the feedback is considered and analysed
Presenting	Present information using: Sounds, Pictures, Words, Video	The presented information has clear connections across formats or ideas	Presentation (or explanation of presentation) has explicit conceptual underpinning	The presentation, methods and results are critiqued and evaluated	Critiqued and developed ideas or new knowledge is presented.	
Processing information	Information is processed, or data/images are manipulated in isolation	Connections are made between or within processes, information/data or images and relevant concepts	Processed data or information has clear conceptual underpinning	Process and product are critiqued and evaluated	Ideas and new knowledge are developed	
Gaming or interactive programmes	Play a game; Take a quiz; Enter a virtual world	Links made between the game/quiz/virtual world and other knowledge	The relevant concepts within the game, quiz or virtual world are identified and explained.	The game, quiz or virtual world is critiqued and evaluated within a conceptual context	Original ideas are used to create a knowledge product in any medium	
Communicating	Send a communication; Receive a communication; Read a communication	Ideas compared and shared with other learners through a two-way conversation (written or verbal)	Communication explicitly develops conceptual understanding	Critique other peoples' work or ideas	Through interaction and communication new knowledge is constructed	

Table 3
The Level Structure of AI Competency Framework (Cukurova & Miao, 2024, p. 22).

Aspects	Progression		
	Acquire	Deepen	Create
1. **Human-centered mindset**	Human agency	Human accountability	Social responsibility
2. **Ethics of AI**	Ethical principles	Safe and responsible use	Co-creating ethical rules
3. **AI foundations and applications**	Basic AI techniques and applications	Application skills	Creating with AI
4. **AI pedagogy**	AI-assisted teaching	AI–pedagogy integration	AI-enhanced pedagogical transformation
5. **AI for professional development**	AI enabling lifelong professional learning	AI to enhance organizational learning	AI to support professional development

model" capable of serving as an external trigger for the development of learners' mental models, thereby positioning AI as a collaborative cognitive agent rather than a passive tool.

These emerging models significantly advance the conceptual understanding of how AI can interact with and scaffold human learning processes, and what knowledge and skill sets are expected from teaching in the era of emerging technologies and AI. They highlight the potential for AI not only to automate routine tasks but also to augment human cognitive capacities and enable new forms of collaborative learning. Nevertheless, their highly theoretical and abstract nature poses challenges for practical implementation in educational settings. They require careful contextualization to translate their conceptual insights into tangible instructional strategies, assessment designs, and curriculum models that meaningfully integrate GenAI while maintaining human agency and fostering critical, ethical, and creative engagement.

2.2.5. Limitations of existing competency frameworks

Frameworks such as TPACK and DigCompEdu have been widely adopted in early research exploring teacher competencies in the AI era (Celik, 2023; Kim et al., 2021; Mishra et al., 2023; Misthou & Paliouras, 2021; Quast et al., 2023; Sun et al., 2023). Their strengths lie in providing structured models for understanding the intersection of technology, pedagogy, and content knowledge, as well as establishing baseline expectations for digital proficiency in educational practice (Celik, 2023; Mishra et al., 2023). However, when considered in relation to the distinctive demands of GenAI-enabled higher education, several significant limitations become apparent.

First, much of the existing research and framework development has focused on the K-12 sector, with comparatively less attention to the unique complexities of university teaching (Kohnke et al., 2023). Higher education brings a distinct set of challenges, including greater student autonomy, disciplinary diversity, and more sophisticated cognitive development, which are not fully addressed in current frameworks (Ng et al., 2023). Second, while technical knowledge and the operationalization of AI tools are well represented, there is often insufficient focus on the pedagogical strategies required to guide, scaffold, and empower students' engagement with GenAI (Cha et al., 2024). This narrow emphasis risks positioning AI integration as a technical exercise rather than as a catalyst for transformative pedagogy that fosters higher-order thinking, creativity, and ethical reasoning. Third, most existing studies are predominantly framed from the teachers' perspective, overlooking the crucial role of students' experiences and perceptions in evaluating the effectiveness and relevance of AI-integrated teaching practices (Alan & Güven, 2022; Bojukrapan et al., 2023). Lastly, while frameworks such as the UNESCO AI Competency Framework emphasize ethical considerations and human-centered approaches, they often remain at a policy or conceptual level, providing limited actionable guidance for discipline-specific and context-responsive pedagogical design (Cheng et al., 2024). Similarly, frameworks such as HCAI, AIED-HCD, and Starkey's Digital Age Learning Matrix, while offering valuable theoretical lenses, require significant contextual adaptation to address the dynamic, co-evolving relationship between students and GenAI systems in contemporary university classrooms.

Taken together, these observations suggest that while existing frameworks offer important foundations, they do not yet fully capture the breadth and complexity required for GenAI-responsive teaching in universities. There is a clear need for a more holistic competency framework that integrates technical, pedagogical, ethical, and developmental dimensions, foregrounds both teacher and student perspectives, and is adaptable to the diverse realities of higher education. Such a framework should articulate not only what university teachers need to know and do, but also how they can foster students' critical, creative, and responsible engagement with GenAI, thereby preparing learners for the demands of an AI-mediated future.

3. Systematic development of the competency framework

3.1. Theoretical foundations

The GenAI-Responsive Competency Framework for University Teachers is built upon an integrated set of theoretical perspectives that collectively articulate the evolving roles of GenAI, teachers, and students in contemporary higher education. These foundations not only justify the framework's structure but also illuminate the pedagogical and social considerations essential for meaningful integration of GenAI.

3.1.1. Constructivist learning theory

Central to the framework is Constructivist Learning Theory, which posits that knowledge is actively constructed by learners through interactions with their environment, tools, and social contexts (Bada & Olusegun, 2015; Woo & Reeves, 2007). Within this paradigm, GenAI is reconceptualized from a mere instructional aid to a dynamic cognitive partner, thinking collaborator, and learning companion (Zhang & Wang, 2025). It facilitates iterative knowledge construction, supports conceptual development, and promotes critical engagement, thereby enhancing students' metacognitive skills and higher-order thinking (Chen & Zhu, 2023; Tran et al., 2025).

3.1.2. Student-centered philosophy

Aligned with constructivism, the framework embraces student-centered learning philosophies that position learners as active agents in the co-creation of knowledge (Abdullah, 2020; Damşa et al., 2019). In this model, GenAI serves as a facilitative agent rather than a passive content delivery mechanism. It empowers students to direct their inquiry, formulate reflective questions, and engage in dialogic exploration of complex ideas (Tang & Putra, 2025). Under the guided facilitation of teachers, learners are encouraged to critically evaluate GenAI outputs, test hypotheses, and externalize their reasoning, fostering intellectual autonomy and agential participation (Bai & Wang, 2025).

3.1.3. Sociocultural perspective

The framework further incorporates sociocultural perspectives, which emphasize that education is embedded within broader institutional, cultural, and social systems (Palincsar, 1998). GenAI integration is thus understood as a socially situated practice, necessitating critical attention to issues of equity, ethics, accessibility, and agency (Jin et al., 2025; Lan & Chen, 2024). This lens ensures that GenAI is implemented in ways that reinforce democratic and inclusive educational values, positioning technology as an enabler of equitable participation and socially responsible innovation.

3.1.4. Learner–GenAI interaction

Building on these foundations, the framework identifies interaction between learners and GenAI as a defining characteristic of education in the AI era (Law, 2024). In contemporary learning environments, students engage in real-time, dialogic exchanges with AI systems capable of generating, adapting, and responding intelligently to inputs (Zhai & Wibowo, 2023). This interaction represents a shift from traditional transmissive models to a collaborative knowledge-building process in which machine intelligence serves as an active co-participant.

Through such interactions, learners are prompted to critique, question, adapt, and synthesize information, thereby cultivating higher-order cognitive competencies and metacognitive awareness (Borge et al., 2024; Chan & Colloton, 2024). Rather than passively accepting AI-generated content, students learn to assess its validity, relevance, and ethical implications (Lee & Low, 2024; Lee & Song, 2024). This iterative process nurtures critical thinking, creative problem-solving, and responsible innovation, positioning learners as co-agents alongside GenAI in the construction of knowledge.

Guided by these theoretical principles, the framework calls for teachers to intentionally design learning experiences and environments that foster critical, creative, and ethical engagement between students and GenAI. It highlights the need for pedagogical strategies that leverage GenAI's capabilities while ensuring that its use aligns with foundational educational goals and humanistic values.

3.2. Theoretical mapping of the framework

The connections between these theoretical foundations and the four competency dimensions are synthesized in Table 4, which illustrates how each theory distinctly informs the framework's structure and focus.

3.3. Systematic development and validation process of the framework

To ensure both theoretical robustness and practical relevance, the GenAI-responsive competency framework was developed through a systematic, multi-stage process grounded in established qualitative research protocols (Khatri et al., 2026). This process combined comprehensive literature engagement with expert consultation, iterative analysis, and consensus-driven refinement.

3.3.1. Expert panel composition

A purposively selected international expert panel was assembled to capture a broad range of perspectives relevant to GenAI integration in higher education. The panel comprised eight members: four educational technology specialists with expertise spanning learning analytics, curriculum reform, assessment innovation, quality assurance, and faculty development; three disciplinary experts from computing education and teacher education; and one university teacher with practical experience in GenAI implementation. This composition was intentionally designed to ensure that both theoretical and applied dimensions were thoroughly represented throughout the framework's development.

3.3.2. Data collection procedures

The development process began with an extensive literature review and comparative analysis of existing competency models, which informed the initial dimensions of the framework. Data collection proceeded through a series of seven structured consultation sessions with the expert panel, conducted between October 2024 and July 2025. Each session was guided by a detailed agenda addressing specific aspects of the framework, such as scope, dimensionality, terminology, and indicator development. Throughout these sessions, panelists engaged in open-ended discussion, provided written and verbal feedback, and contributed directly to the iterative refinement of framework

Table 4
Mapping of theoretical foundations to framework dimensions.

Theoretical Foundation	Core Tenets & Relevance to GenAI	Implications for Framework Dimensions
Constructivist Learning Theory (3.1.1)	Knowledge is actively constructed by learners through interaction. GenAI serves as a cognitive partner and thinking collaborator to facilitate knowledge construction and critical engagement.	• Informs all dimensions by positioning GenAI as a mediator of learning. • Especially critical for: Teaching & Learning (designing interactive, co-constructive activities); Assessment (promoting reflective, generative evaluation).
Student-Centered Philosophy (3.1.2)	Learners are active agents. GenAI acts as a facilitative agent to empower student inquiry, dialogue, and intellectual autonomy.	• Informs all dimensions by centering on student agency. • Especially critical for: Curriculum/Learning Design (personalizing pathways); Teaching & Learning (fostering student-led exploration).
Sociocultural Perspective (3.1.3)	Education is a social practice situated within cultural and institutional contexts. GenAI integration must address equity, ethics, accessibility, and social responsibility.	• Informs all dimensions by providing an ethical and contextual lens. • Especially critical for: GenAI Literacy (critical evaluation of bias, ethical use); Assessment (ensuring fairness, inclusivity).
Learner-GenAI Interaction Paradigm (3.1.4)	Defines the new triadic dynamic of Teacher–Student–GenAI collaboration. Emphasizes dialogic exchange and co-creation of knowledge with an agentic AI.	• Informs all dimensions by redefining roles and interactions. • The primary rationale for the framework's dual-perspective (Teacher & Student) structure across all dimensions and tiers.

documents. All sessions were meticulously documented, with comprehensive minutes and audio recordings transcribed to ensure accuracy and facilitate subsequent analysis.

3.3.3. *Data analysis and iterative refinement*

Data analysis was conducted using an iterative, consensus-building approach consistent with the Delphi method (Oxley et al., 2025). The validation of the GenAI-responsive competency framework was achieved through seven structured expert consultation sessions, each contributing to the framework's conceptual and operational refinement. The process was systematically documented to ensure transparency and traceability. Table 5 provides a summary of the key issues discussed, actions taken, consensus outcomes, and illustrative quotations from the expert panel during each session.

Transcribed feedback and session notes were subjected to thematic coding to identify recurring areas of agreement, points of contention, and emerging ambiguities (Gawlik, 2018). Key issues, including distinctions between GenAI literacy and competency, considerations for global and disciplinary applicability, and the development of practical indicators, were prioritized for focused discussion in subsequent sessions. After each round of consultation, the framework was revised to incorporate the synthesized input from the panel. Revised drafts were circulated for additional feedback, and this cycle continued until the expert panel reached consensus on all framework dimensions and indicators. Particular attention was paid to ensuring that the framework was relevant across disciplines, aligned with learning sciences, and grounded in actionable practice.

4. A four-dimensional, three-tiered GenAI-responsive competency framework for university teachers

This framework is structured around four core dimensions (i.e., *GenAI Literacy*, *Curriculum/Learning Design*, *Teaching and Learning*, and *Assessment*), and three developmental tiers (i.e., *Basic*, *Intermediate*, and *Advanced*) in Table 6. It introduces an innovative dual-perspective approach, integrating both teachers' own competencies and their capacity to empower students. This structure bridges teacher development with student growth in GenAI-enabled learning environments. By adopting this dual focus, the framework transcends conventional, one-

Table 5
Validation outcomes of the competency framework.

Session	Key Issues/Discussion Points	Actions Taken	Outcomes	Example Quotes from Panelists
1	Ambiguity between the concepts of 'GenAI literacy' and 'competency'. Need to clarify framework scope and ensure cross-disciplinary relevance.	Clarified and consistently used the term 'competency'. Established a multi-dimensional structure focused on GenAI.	Agreement to proceed with clarified dimensions	"We need to differentiate between AI literacy and competency, and be consistent in our use of terms throughout."
2	Need for a focus on comprehensive teacher competencies beyond basic knowledge. Importance of self-assessment for implementation.	Adopted 'competency' as the central construct. Planned for the development of subjective self-assessment tools. Linked the framework to real teaching scenarios.	Consensus reached on focusing on actionable teacher competencies	"The framework should emphasize teachers' comprehensive abilities, not just basic understanding."
3	Overlap and confusion between framework dimensions, such as skill enhancer and collaborative partner. Unclear global relevance.	Revised and reorganized dimensions for conceptual clarity. Removed research and service categories. Expanded the framework for international applicability.	General agreement on revised structure	"We must ensure the framework is relevant beyond the Chinese context and focused on generative AI."
4	Overlapping subdimensions and lack of clarity in indicators. Insufficient integration of pedagogy and learning science.	Reclassified overlapping elements. Integrated pedagogical and learning science perspectives. Drafted clear indicators to support practical implementation.	Consensus on operationalized and distinct dimensions	"Let's establish indicators and metrics to guide interviews and coding."
5	Lack of specificity in terminology, especially between GenAI and general AI. Insufficient contextualization for higher education.	Standardized all terminology to 'GenAI'. Contextualized examples and guidance specifically for higher education settings.	Agreement on specificity and contextual adaptation	"Highlighting the difference between GenAI use in higher education and K-12 is essential."
6	Unclear hierarchy among dimensions. Conflation of skill automation and collaborative partnership. Reflection and growth dimension was overly broad.	Developed a hierarchical matrix. Differentiated skill automation from collaborative partnership. Refined and focused the reflection and growth dimensions.	Agreement on revised matrix and clarity among dimensions	"How do we show the hierarchical relations between each dimension clearly in the framework?"
7	Final validation of overall framework coherence and indicators. Assessment of readiness for empirical piloting.	Confirmed all framework dimensions and indicators. Finalized and validated the interview protocol that is linked to the framework.	Unanimous endorsement of the framework for use	"The preliminary framework will be updated in progress through interviews. This is a living instrument."

Table 6
Proposed theoretical framework.

Competency Dimension	Competency levels		
GenAI Literacy	**Basic**	**Intermediate**	**Advanced**
Teacher's own literacy	Demonstrates foundational understanding of safe, ethical, and effective interaction with GenAI systems.	Critically engages with GenAI outputs, identifying potential issues, biases, and opportunities for improvement.	Explores innovative ways of collaborating with GenAI, reflecting on its evolving role as a cognitive partner.
Empowering students' literacy	Introduces students to responsible, ethical, and basic methods for interacting with GenAI systems.	Guides students to critically engage in iterative dialogues with GenAI tools.	Supports students in co-creating knowledge and artefacts through advanced human-AI collaboration.
Curriculum/ Learning Design	**Basic**	**Intermediate**	**Advanced**
Teacher's own usage	Uses GenAI to generate ideas and curate teaching resources, recognizing its potential for curriculum innovation.	Collaborates with GenAI to refine, adapt, and personalize curricular materials responsively.	Co-designs interdisciplinary, project-based curricula that integrate student-GenAI partnership models.
Empowering students' usage	Introduces students to GenAI-supported curriculum exploration and learning design.	Facilitates students' collaborative construction of learning artefacts using GenAI tools.	Empowers students to co-design meaningful, GenAI-enhanced, self-directed learning experiences.
Teaching and Learning	**Basic**	**Intermediate**	**Advanced**
Teacher's own usage	Uses GenAI to support basic instructional tasks, such as generating examples, explaining concepts, or developing simple interactive materials.	Applies GenAI for dynamic scaffolding, simulations, or adaptive feedback to personalize student learning pathways.	Designs and facilitates co-learning environments where GenAI serves as a cognitive partner in inquiry, co-construction, and classroom dialogue.
Empowering students' usage	Introduces students to using GenAI for basic exploration, such as brainstorming, retrieving information, or generating explanations.	Guides students to interact critically and collaboratively solve problems with the support of GenAI.	Enables students to co-create knowledge through GenAI-supported innovation, reflective dialogue, and the development of creative projects.
Assessment	**Basic**	**Intermediate**	**Advanced**
Teacher's own usage	Uses GenAI to automate simple assessments and provide immediate feedback.	Adopts GenAI to design adaptive formative assessments that are responsive to students' learning trajectories.	Co-creates innovative, AI-enhanced assessment models that integrate both human and machine feedback for deeper learning insights.
Empowering students' usage	Introduces students to interactive self-assessment tools powered by GenAI.	Guides students in interpreting and applying GenAI-generated feedback to enhance their work.	Support students in designing AI-enhanced peer and self-assessment systems, promoting reflective and collaborative evaluation processes.

directional models that address either teachers or students alone, reflecting a more holistic and interactive approach to competency development in the era of GenAI.

By placing equal emphasis on the teacher's role in empowering students, the framework embodies the educational principle that teacher development directly advances student learning. The framework also captures the evolving relationship between teachers and students in GenAI-mediated environments, shifting from a traditional dyadic instructional model to a triadic dynamic of collaborative exploration among teachers, students, and GenAI (Prilop et al., 2025). This shift is exemplified through an emphasis on “co-design” and “co-creation” at the *Advanced* tier. Additionally, the three-dimensional matrix (Dimensions × Tiers × Perspectives) provides a clear developmental pathway, offering teachers a visual and actionable roadmap for self-assessment and professional growth (Fig. 2). By embedding critical thinking, ethical reasoning, and responsible use throughout all dimensions and tiers, the framework directly addresses contemporary challenges posed by GenAI, such as bias and academic integrity, aiming to develop teachers and students who can both adapt to and shape the future of AI in education (Salido et al., 2025).

The selection of the four dimensions: *GenAI Literacy*, *Curriculum/ Learning Design*, *Teaching and Learning*, and *Assessment*, is grounded in the theoretical foundations discussed earlier and reflects a deliberate focus on the core domains of educational practice: curriculum, instruction, and evaluation (Matsumoto-Royo & Ramírez-Montoya, 2021). Each dimension captures a distinct facet of how GenAI transforms teaching and learning, while also accounting for the evolving roles of both teachers and students within GenAI-enhanced environments. These dimensions are not isolated; rather, they interact dynamically and are deeply intertwined with the development of both teacher and student GenAI literacy.

As Table 6 *shows, GenAI Literacy* establishes the foundational competence for meaningful engagement with AI technologies, shaping how teachers and students approach all other domains (Tadimalla & Maher, 2025). It is crucial to clarify that, while the proposed framework also incorporates GenAI Literacy as a core dimension, just like existing ones, our conception fundamentally differs from the narrow, technically focused “AI literacy” of prior models. We redefine it as a critical and applied form of literacy that integrates ethical reasoning, pedagogical judgment, and contextual awareness, serving as the essential foundation upon which the other pedagogical dimensions are built, rather than an end in itself.

Curriculum/Learning Design focuses on teachers' competency to search, identify, synthesize, and design what and how GenAI can be incorporated into the curriculum or learning process through pedagogical strategies, learning activities and materials, and pathways prior to actual teaching and learning scenarios (Avcı, 2026; Cheng et al., 2024). The *Teaching and Learning* dimension emphasizes how the curriculum/learning design is delivered and how students engage with it during the teaching and learning scenarios. This addresses the transformational shift in pedagogical methods and relationships facilitated by GenAI, transitioning from a transmission-based approach to the co-construction of knowledge (Otto et al., 2025). Finally, the *Assessment* dimension highlights how GenAI enables more adaptive, responsive, and participatory evaluation practices.

Across all dimensions, GenAI's role evolves from a basic tool for automation or support to a collaborative partner capable of critical and

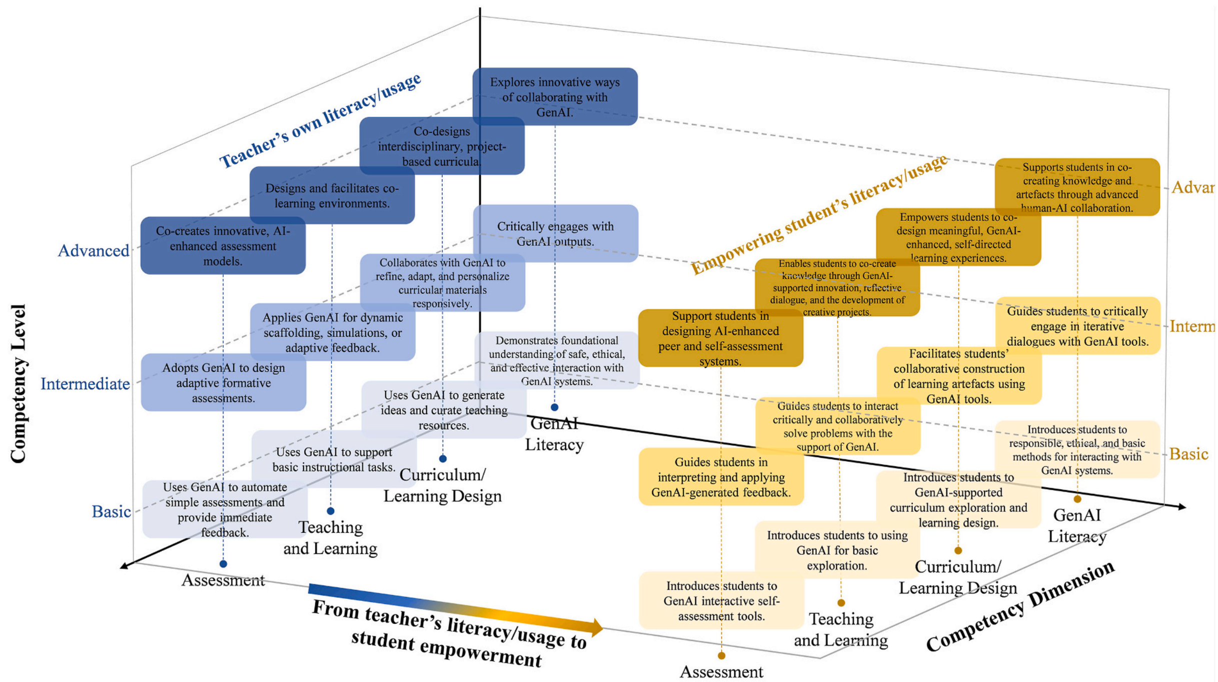

Fig. 2. GenAI responsive competency framework for university teachers.

creative engagement. Horizontally, the framework organizes university teachers' competency development across three progressive levels: Basic, Intermediate, and Advanced. The progression establishes benchmarks for teacher competency, defining a developmental trajectory that underscores the synergistic relationship among four core dimensions (Cukurova & Miao, 2024; Song et al., 2025). At the Basic Level, teachers build awareness and operational competence. They use GenAI for routine tasks such as generating ideas, creating simple materials, or automating low-stakes assessments. They begin to incorporate ethical and safety considerations into their practice. In parallel, they introduce students to GenAI tools, model responsible use, and encourage learners to use AI-generated content to explore concepts or conduct self-assessment. The emphasis is on familiarity and ethically grounded experimentation.

At the *Intermediate Level*, teachers transition to guided application and critical engagement. They use GenAI not only for automation but also as an assistant to refine teaching resources, design adaptive assessments, and support personalized learning (Naznin et al., 2025). A key advance is their growing competency to critically evaluate AI outputs, including detecting bias and assessing pedagogical appropriateness. Correspondingly, they guide students to use GenAI dialogically, prompting iteratively, co-creating artefacts, interpreting GenAI feedback, and questioning the reliability and relevance of generated content. Critical thinking and reflective interaction are central at this level.

The *Advanced Level* represents a creative and epistemic partnership with GenAI. Teachers treat GenAI as a cognitive collaborator in co-designing interdisciplinary projects, facilitating dynamic learning environments, and building human-AI feedback loops into assessment (Memon & Kwan, 2025). They innovate pedagogically and reflect continuously on the evolving relationship between human and machine intelligence. Students, in turn, are supported in becoming co-creators and critical agents, using GenAI to tackle complex problems, produce original work, and thoroughly examine the social, ethical, and technical dimensions of AI (Chee et al., 2025; Michel-Villarreal et al., 2023). The educational goal shifts to transformative learning, integrating critical, creative, and ethical capabilities with advanced AI literacy.

These levels represent a teacher's evolving capability, from foundational awareness and operational proficiency, through guided application and critical integration, to creative partnership and knowledge co-creation with GenAI. Rather than constituting a rigid hierarchy, these levels serve as developmental benchmarks that acknowledge the contextual and dynamic nature of teacher learning and practice. Together, the four dimensions and three levels form a comprehensive matrix that emphasizes the fostering of student autonomy, digital judgment, and collaborative agency across all GenAI-mediated educational activities. The framework provides a dynamic and holistic model for advancing teacher readiness and pedagogical innovation in the GenAI era. The following section offers a detailed description of each dimension and developmental level within the proposed framework.

4.1. GenAI literacy across levels

The *GenAI Literacy* dimension, as outlined in Appendix A, offers a structured and developmental progression of teacher competencies necessary for meaningful engagement with GenAI. This progression moves from operational and safety-aware interactions to critical, reflective, and ultimately co-creative uses of GenAI in a broad way.

At the *Basic Level*, teacher competency focuses on functional access and ethical caution. Teachers are expected to navigate platforms, write clear prompts, and recognize core limitations, such as potential inaccuracies and data privacy concerns (Francis et al., 2025). Empowering students at this level involves introducing them to the tool's functionality, effective prompting strategies, and foundational ethical guidelines (Karaoglan Yilmaz, Yilmaz, & Ceylan, 2024). This level establishes a baseline of safe and operational proficiency for both teachers and learners.

Advancing to the *Intermediate Level*, the focus shifts to critical engagement and contextual adaptation. Teachers develop skills to evaluate outputs for accuracy, bias, and relevance, refine prompts, and supplement responses as needed. They also reflect on ethical implications with colleagues and students. In parallel, students are guided to engage in iterative dialogue with GenAI, posing follow-up questions and critically examining the outputs for errors and biases. This level cultivates a discerning and dialogic approach to human-AI interaction.

The *Advanced Level* centers on innovation, co-creation, and reflective collaboration. Teachers are envisioned as designers of new workflows that integrate GenAI as a “co-thinker” and actively contribute to professional communities by sharing insights on human-AI partnerships. Similarly, students undertake self-directed projects where GenAI serves as a collaborative partner in generating original work, such as research reports or multimedia presentations, while critically reflecting on both the process and the AI's role. This level aims to foster a transformative and generative relationship between learners and AI.

4.2. Curriculum/learning design across levels

The second dimension, *Curriculum/Learning Design* (Appendix B), addresses how GenAI can be utilized to plan, develop, and adapt curricular content, instructional materials, and learning activities in alignment with intended learning outcomes (Cheng et al., 2024; Kim, 2025). This dimension encompasses both teachers' ability to leverage GenAI in instructional design and their capacity to empower students to use GenAI in shaping their own learning experiences.

At the Basic Level, teacher competency focuses on supplementing and enriching existing curriculum structures. Teachers use GenAI to generate initial ideas, compile teaching resources, and test AI-supported lesson preparation, without fundamentally altering the prescribed curriculum. In parallel, students are introduced to GenAI as a tool for exploring topics, generating examples, and stimulating inquiry within teacher-guided boundaries.

The Intermediate Level shifts to adaptation and personalization. Teachers utilize GenAI to tailor materials and assignments based on learner differences, such as their level of prior knowledge, characteristics, and performance feedback, iteratively refining AI-generated resources. Students are supported in using GenAI collaboratively by co-creating presentations and reports, refining AI-generated drafts, and critically overseeing GenAI's contributions in group projects.

At the Advanced Level, the focus is on co-creation and participatory design. Teachers collaborate with students and GenAI to develop interdisciplinary, project-based modules, using AI to facilitate real-world data analysis and interpretation within pedagogically structured frameworks. Students are empowered to co-design personalized learning pathways—selecting themes, resources, and pacing with GenAI guidance, while reflecting on their progress and aligning self-directed goals with curricular or their own learning expectations.

4.3. Teaching and learning across levels

The *Teaching and Learning* dimension addresses the integration of GenAI into instructional practices to foster interactive, adaptive, and collaborative learning environments (Appendix C). It emphasizes the use of GenAI to support personalized instruction, provide dynamic feedback, and scaffold student learning (Ba et al., 2025). A central aspect of this dimension is empowering students to collaborate with GenAI in problem-solving, creative tasks, and reflective inquiry, thereby consciously developing their metacognitive awareness and agency throughout the learning process.

At the Basic Level, teacher practice is characterized by supplementing and automating conventional instruction. Teachers use GenAI to generate examples, create simple interactive materials, and automate routine tasks such as drafting discussion prompts, while reviewing AI outputs for accuracy (Xia et al., 2024). Students are introduced to GenAI as an informational and explanatory tool, engaging in basic exploration activities, such as querying and summarizing, under the guidance of their teachers.

The Intermediate Level shifts toward adaptive scaffolding and dialogic engagement. Teachers employ GenAI to offer real-time personalized hints and feedback, adjust activity complexity based on student responses, and tailor instruction using AI-generated performance insights. Students are coached to use GenAI iteratively, refining solutions, evaluating feedback, and collaborating with both AI and peers in problem-solving contexts.

At the Advanced Level, teaching and learning evolve into co-constructive and critically reflective practices. Teachers implement the designed activities where GenAI serves as an active participant in exploring complex problems, facilitating brainstorming, and supporting reflective discourse during teaching and learning. Students are empowered to collaborate dynamically with GenAI, generating innovative ideas, critically evaluating AI input alongside peer perspectives, and iteratively refining artefacts through sustained interaction within the learning environment that teachers created for them. This empowerment will not be restricted to the classroom, but also exerts a positive influence on students' self-directed learning beyond the curriculum designed by teachers in the long run (H. Kim et al., 2025).

4.4. Assessment across levels

The *Assessment* dimension focuses on the use of GenAI to design, deliver, and interpret both formative and summative evaluations (Appendix D). This includes automating routine assessment tasks, generating adaptive and personalized feedback, and developing innovative evaluation systems that integrate and highlight AI insights with human judgment (Alsalem, 2024; Chatterjee & Bhattacharjee, 2020). Importantly, this dimension also emphasizes empowering students to use GenAI for self-assessment, interpret AI-generated feedback, and actively participate in designing peer and self-evaluation processes.

At the Basic Level, teacher practice centers on automation and standardization. Teachers use GenAI to automatically grade quizzes and assignments, generate basic assessment questions aligned with learning objectives, and provide immediate feedback (Moorhouse et al., 2023). Students are introduced to GenAI-supported self-evaluation through practice exercises and quizzes, enabling them to independently check their understanding of course material and their learning outcomes.

The Intermediate Level emphasizes adaptation and progress monitoring. Teachers employ GenAI to design formative assessments that adjust in difficulty based on student performance, track learning progress over time, and tailor assessment tasks accordingly. Students are encouraged to learn to interpret and apply GenAI-generated feedback to revise and refine their work, and use GenAI tools to monitor their own progress and identify areas for improvement and growth consciously.

At the Advanced Level, assessment evolves into collaborative and reflective evaluation. Teachers collaborate with GenAI to develop innovative assessment models that combine machine analytics with human judgment, creating multi-dimensional feedback processes that integrate AI insights with personalized comments (Alsalem, 2024; Luckin et al., 2022). Students are empowered to design and implement GenAI-enhanced peer and self-assessment frameworks, engage in reflective evaluation processes that analyze both AI and human feedback, and develop metacognitive awareness by critically assessing different sources of feedback.

4.5. Illustrative application of the framework

To illustrate the application of the proposed framework in practice, Appendix E presents a worked example from a postgraduate Research Methods course in which students develop a research proposal. The example shows how teachers' GenAI Literacy progresses from safe operational use (Basic), to critical evaluation and contextual adaptation (Intermediate), and then to reflective co-creation with GenAI as a “co-thinker” (Advanced), while simultaneously empowering students to engage in iterative prompting, verification, and ethical boundary-setting. It also illustrates how Curriculum/Learning Design, Teaching and Learning, and Assessment evolve from using GenAI to supplement resources and automate low-stakes tasks, to designing adaptive supports and dialogic learning activities, and ultimately to co-designed learning experiences and collaborative evaluation practices that synthesize teacher judgment, peer review, and AI-informed feedback.

5. Conclusions, discussions, and implications

The four-dimensional, three-tiered GenAI-Responsive Competency Framework presented in this paper offers a comprehensive approach to preparing university teachers for the challenges and opportunities presented by GenAI in higher education. By emphasizing the reciprocal relationship between teacher development and student empowerment, and by positioning GenAI as both a tool and a collaborative partner, the framework lays a foundation for more innovative, inclusive, and future-ready educational practices. Its successful implementation will require ongoing refinement, cross-institutional collaboration, and sustained engagement with key educational stakeholders. Ultimately, this work contributes to the advancement of educational ecosystems where teachers and students can thrive together as critical, creative, and ethical participants in an AI-mediated world.

5.1. Implementation challenges and contextual considerations

While the proposed framework offers a visionary pathway for GenAI integration, its implementation faces significant practical challenges that must be acknowledged and addressed. First, institutional readiness varies considerably across higher education contexts. Many institutions lack clear GenAI usage policies, robust infrastructure, and sustained professional development support, creating implementation barriers that cannot be overcome through individual teacher effort alone (Francis et al., 2025; Nguyen, 2025). Second, issues of digital equity and access persist. Not all students or educators have equal access to premium GenAI tools or the digital literacy necessary to use them effectively, which risks exacerbating existing educational disparities (Kohnke et al., 2023). Third, the framework's effectiveness may be limited in contexts where GenAI adoption is nonexistent or minimal, suggesting the need for adaptable implementation strategies that account for varying levels of technological adoption.

To address these challenges, we recommend adaptive implementation approaches that include: (1) developing tiered support systems that accommodate different institutional readiness levels; (2) creating open-access GenAI resources and digital literacy programs to promote equitable access; and (3) establishing clear guidelines for ethical and inclusive GenAI use that address potential biases in AI systems. Successful implementation will require coordinated efforts among institutional leaders, policymakers, and educators to create the necessary enabling conditions for this framework to achieve its transformative potential.

It is important to consider the framework's applicability in contexts where GenAI adoption remains limited or entirely absent. While designed to be GenAI-responsive, the framework's developmental architecture ensures its relevance even in such environments. The “Basic” tier serves as a vital entry point for building foundational awareness and literacy, providing a structured pathway for educators and institutions at pre-adoption stages to commence their developmental journey. Competencies at this initial level focus on understanding GenAI's fundamental capabilities and limitations, examining its ethical implications, and considering its potential pedagogical relevance, all essential preparatory steps requiring no immediate tool usage. Consequently, rather than constituting a rigid assessment instrument presuming existing implementation, the framework functions more broadly as a developmental roadmap. It is designed to scaffold progression from foundational understanding toward practical application, rendering it a valuable tool for strategic planning and professional development even in contexts where GenAI integration remains prospective.

This developmental approach acknowledges the varied adoption landscapes across higher education while maintaining the framework's forward-looking orientation. The tiered structure allows institutions to identify appropriate starting points based on their current technological readiness, with the Basic tier providing an essential foundation-building for those yet to embark on GenAI integration. This flexibility ensures the framework's relevance across diverse implementation contexts, from early exploration to advanced stages of adoption.

5.2. Implications for the scholarship of higher education Teaching and learning

The framework moves beyond instrumental approaches to educational AI by advancing a dynamic co-evolutionary model of teacher-student-AI development across dimensions of GenAI literacy, curriculum/learning design, teaching and learning, and assessment. It reconceptualizes GenAI as a cognitive partner and reflective catalyst, rather than merely a productivity tool, thereby enabling a paradigm shift from static competency frameworks to ecological understandings of learning in AI-pervasive environments. The framework captures how teacher competencies evolve through continuous interaction with both students and AI systems, creating a feedback loop where pedagogical practices are refined based on AI-mediated teaching experiences and student responses.

This study establishes a new research agenda by proposing a triadic teacher–student–GenAI research paradigm that examines the complex interactions and mutual adaptations among these three roles. The framework offers measurable constructs for examining how teacher competencies evolve through AI-mediated practices, how student learning behaviors impact teacher growth, and how GenAI serves as both a tool and a catalyst in this process. It opens new methodological pathways for studying the co-evolution of human and AI in educational settings.

Implementation of this framework requires institutional support for professional learning communities, enabling teachers to share and reflect on their experiences with GenAI-mediated teaching. Future research should employ longitudinal designs to track the reciprocal development of teachers and students in GenAI-enhanced environments, with a particular focus on how teachers adapt their practices in response to AI-generated feedback and student interactions with technology.

5.3. Implications for higher education Teaching and learning practices and policies

The framework provides higher education with a transformative roadmap for AI-integrated teaching and learning. It enables a shift from teacher-centered knowledge delivery to **c**o-created learning experiences where teachers and students collaboratively engage with GenAI as partners in knowledge construction. The four-dimensional structure supports discipline-sensitive implementation across academic contexts, while maintaining core principles of ethical integration, pedagogical innovation, and quality enhancement.

A key innovation of the framework lies in its recognition of reflective practice through AI-mediated teaching. Teacher competencies are developed dynamically as teachers observe how students interact with GenAI, analyze AI-generated insights, and adapt their approaches accordingly. This creates an ongoing cycle of pedagogical improvement, where teacher development, student engagement, and AI capabilities evolve in tandem.

5.4. Practical implications

For educational institutions and policymakers, this framework provides an actionable roadmap for supporting the integration of GenAI through four strategic areas: faculty development, curriculum design, assessment reform, and institutional policy-making. The developmental progression across three tiers (Basic, Intermediate, Advanced) enables institutions to design targeted professional development programs that meet teachers at their current competency levels while providing clear pathways for growth. Specifically, we recommend.

1) Developing GenAI-focused training modules aligned with the framework's dimensions and levels;

2) Creating self-assessment instruments to help teachers identify competency gaps and track progress;
3) Establishing communities of practice where university teachers can share experiences and strategies for GenAI integration;
4) Revising curriculum guidelines and assessment policies to support ethical and pedagogically sound GenAI use.

Successful implementation requires recognizing that GenAI competency development occurs within broader institutional and systemic contexts. Key enablers include institutional support structures, quality professional development opportunities, flexible curriculum policies, and teacher motivation. Rather than treating these as internal dimensions of the framework, we position them as essential external factors that significantly influence the development of competencies in practice (Sancar et al., 2021). This distinction acknowledges that effective GenAI integration requires both individual capability building and supportive organizational conditions.

By offering a structured, developmentally responsive competency model, this study will empower teachers to design GenAI-integrated curriculum and learning activities aligned with students' cognitive development and disciplinary needs. It will provide practical tools and strategies for instructional design and assessment, fostering critical thinking, ethical reasoning, and innovation among students. Furthermore, this research enriches the discourse on AI in education by emphasizing GenAI-responsive pedagogy over mere technical or ethical literacy. It lays the groundwork for future studies on discipline-specific frameworks and longitudinal impacts of GenAI adoption on education quality, advocating for a rethinking of teacher competency frameworks in the GenAI era.

6. Limitations and future research

6.1. Limitations of the study

Notwithstanding its systematic development through literature synthesis and expert validation, the proposed GenAI competency framework presents several conceptual and methodological constraints that merit scholarly acknowledgment. The framework remains predominantly theoretical, with empirical validation confined to preliminary stages. While expert consultations have informed its construction, broader implementation across varied institutional settings is necessary to verify its practical efficacy and establish generalizability. Methodologically, the validation process relied on a limited and geographically concentrated expert cohort. This approach, though valuable for establishing face validity, risks incorporating perspectival biases. Future iterations would be strengthened through engagement with more diverse international stakeholders, particularly including early-career academics and student voices currently underrepresented in the validation process. Furthermore, the framework's specific orientation toward higher education contexts potentially restricts its direct applicability to alternative educational environments such as K-12 or vocational training systems. Significant contextual adaptation would be required for meaningful implementation beyond university settings, suggesting the need for subsequent domain-specific modifications. These limitations collectively highlight the framework's provisional character while delineating productive pathways for its continued scholarly refinement and practical development.

6.2. Future directions

Building upon the proposed GenAI-responsive competency framework, future work should focus on translating theoretical dimensions into actionable tools through three key initiatives.

The immediate priority involves developing comprehensive self-assessment instruments that operationalize the framework's four competency dimensions across proficiency levels. These tools will enable educators to systematically evaluate their GenAI competencies, identify developmental gaps, and chart personalized growth pathways. Following instrument development, rigorous validation across diverse institutional contexts becomes essential. Mixed-methods approaches combining quantitative techniques—particularly Confirmatory Factor Analysis (CFA) and Structural Equation Modeling (SEM) following established methodologies (Brown & Moore, 2012), with qualitative cross-disciplinary case studies will establish the framework's psychometric robustness while ensuring its practical adaptability across academic fields. Beyond measurement tools, the framework's conceptual architecture requires refinement through explicit recognition of contextual enablers. Professional development opportunities, institutional policy environments, and curriculum support systems should be formally positioned as external factors that actively shape competency development, as evidenced by recent research (Cheng et al., 2024). This reconceptualization aligns with growing scholarly consensus that GenAI competency development represents not merely an individual endeavor but a systemically embedded process, necessitating consideration of both individual skills and the institutional conditions that enable their growth (Celik et al., 2022; Song et al., 2025). The ultimate sustainability of the framework depends on its seamless integration into institutional ecosystems. This necessitates developing GenAI-aligned training modules, embedding competencies into teacher certification programs, and creating digital platforms for continuous professional learning. Such implementations will ensure the framework's lasting impact on educational practices and student outcomes.

Through these coordinated efforts—instrument development and validation, conceptual refinement, and institutional integration—the GenAI competency framework can evolve from a theoretical construct to a transformative tool, effectively preparing higher education for an AI-augmented future.

CRediT authorship contribution statement

Daner Sun: Writing – original draft, Visualization, Methodology, Conceptualization. **Shen Ba:** Writing – original draft, Visualization, Methodology, Conceptualization. **Yingying Cha:** Writing – review & editing, Resources, Project administration, Investigation. **Jiahui Yu:** Writing – review & editing, Resources, Project administration, Investigation. **Feng-Kuang Chiang:** Writing – review & editing, Validation, Supervision. **Hai Min Dai:** Writing – review & editing, Validation, Supervision. **Cher-Ping Lim:** Writing – review & editing, Validation, Supervision.

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Declaration of generative AI in scientific writing

During the preparation of this work, the authors used (EdUHK) ChatGPT and Deepseek to improve the language, readability of this article. After using these tools, the authors reviewed and edited the content as needed and will take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Proposed theoretical framework examples - GenAI literacy

Basic	Intermediate	Advanced
Teacher's own literacy		
• Understands how to access and navigate GenAI platforms. • Practices clear, concise prompt-writing to obtain responses. • Understands that GenAI may produce inaccurate or incomplete information. • Follows privacy and data security rules when using GenAI tools. • Avoids sharing confidential or sensitive information with GenAI systems.	• Reviews GenAI-generated content critically, identifying and verifying inaccuracies or biased language. • Evaluates the relevance and appropriateness of GenAI responses for specific teaching contexts. • Adjusts prompts or supplements GenAI outputs with additional resources to enhance accuracy. • Reflects on the ethical implications of GenAI use and discusses these with colleagues or students.	• Designs and experiments with new workflows that integrate GenAI as a co-thinker in lesson planning or research. • Shares insights on AI-human collaboration in professional learning communities. • Innovates by co-creating teaching materials or research outputs with GenAI assistance.
Empowering students' literacy		
• Introduces students to GenAI tools and basic functionalities. •Shows students how to ask clear and specific questions to the GenAI for accurate answers. •Encourages students to ask questions about GenAI outputs and think about their reliability. • Teaches students the guidelines on privacy and ethical use of GenAI.	• Guides students to engage in back-and-forth conversations with GenAI, asking follow-up questions to clarify and improve GenAI responses. • Encourages students to ask Encourages students to analyze and question GenAI-generated answers, identifying possible errors or biases.	• Facilitates student-centered and self-directed projects where learners and GenAI tools collaboratively generate original research reports, artworks, or multimedia presentations. • Supports students in critically reflecting on GenAI's contributions and limitations throughout the co-creation process. • Encourages students to work with GenAI as a partner for brainstorming, drafting, and refining complex ideas.

Appendix B. Proposed theoretical framework examples - Curriculum/Learning design

Basic	Intermediate	Advanced
Teacher's usage		
• Uses GenAI to generate initial lesson ideas and compile relevant materials. • Selects and organizes GenAI-suggested resources to supplement existing curriculum materials. • Tests GenAI tools to support lesson preparation without changing the overall curriculum structure.	• Uses GenAI to modify lesson content based on student performance data, tailoring materials to different learning levels. • Uses GenAI to create personalized reading lists or assignments that address diverse learner needs. • Adjusts AI-generated curriculum resources iteratively based on student feedback and classroom outcomes.	• Collaborates with students and GenAI to develop project-based learning modules. • Uses GenAI to analyze or interpret real-world data within a structured framework, supported by systematic training of GenAI.
Empowering students' usage		
• Introduces students to GenAI tools that help explore curriculum topics through interactive searches and content generation. • Guides students in using GenAI to find examples, summaries, or explanations related to course themes. • Encourages students to ask questions and investigate topics with GenAI support to spark curiosity and engagement.	• Facilitates group projects where students use GenAI tools to co-create presentations, reports, or digital artefacts. • Guides students in collaboratively refining GenAI-generated drafts, integrating diverse perspectives and feedback. • Supports students in using GenAI to generate ideas and structure content while maintaining critical oversight of GenAI contributions.	• Supports students in co-creating personalized learning pathways and projects by selecting themes, resources, and pacing with GenAI guidance. • Encourages students to plan their learning journey, integrating GenAI-generated insights with personal goals and curricular requirements. • Facilitates reflection on progress and decisions over time within a flexible, GenAI-enhanced curriculum framework.

Appendix C. Proposed theoretical framework examples - Teaching and learning

Basic	Intermediate	Advanced
Teacher's usage		
• Uses GenAI to generate simple examples or explanations to support lesson delivery. • Creates basic interactive materials like quizzes or flashcards with GenAI assistance. • Applies GenAI tools to automate routine instructional tasks, such as drafting discussion prompts. • Reviews GenAI outputs to ensure accuracy before practice.	• Uses GenAI-powered platforms to provide real-time, personalized hints and feedback during problem-solving activities. • Incorporates GenAI-driven activities that adapt complexity based on student responses to scaffold learning effectively. • Analyzes GenAI report on student performance to tailor instruction and support diverse learning needs. • Adjusts teaching strategies dynamically based on GenAI insights into individual student progress.	• Creates learning activities where students explore complex problems with GenAI. • Uses GenAI tools to support real-time brainstorming, idea development, and reflective discussions in class. • Facilitates projects where GenAI helps generate insights and students critically engage with both GenAI and peers. • Encourages co-construction of knowledge by integrating GenAI as an active participant in classroom dialogue.

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Basic	Intermediate	Advanced
Empowering students' usage		
• Introduces students to use GenAI tools to find information and generate explanations during lessons. • Encourages simple GenAI-supported exploration activities, such as asking questions and summarizing content.	• Coaches students to use GenAI tools during specific learning tasks, such as analyzing case studies or conducting experiments, fostering deeper understanding. • Supports students in iterative dialogue with GenAI to refine solutions and critically evaluate GenAI feedback during group work. • Encourages collaborative problem-solving where students and GenAI jointly troubleshoot challenges in real-time learning activities.	• Empowers students to actively collaborate with GenAI during specific projects or tasks, generating innovative ideas and solutions in real time. • Facilitates ongoing reflective dialogue and critical evaluation of GenAI input alongside peer feedback during learning activities. • Supports hands-on creation and iterative refinement of artefacts through sustained student-GenAI interaction within the learning session.

Appendix D. Proposed theoretical framework examples - Assessment

Basic	Intermediate	Advanced
Teacher's own usage		
• Uses GenAI to automatically grade quizzes or assignments and deliver immediate, simple feedback. • Employs GenAI to generate basic assessment questions or prompts aligned with lesson objectives.	• Designs formative assessments that adapt in difficulty or focus based on student responses, supported by GenAI analytics. • Uses GenAI to track student learning progress over time and tailor assessment tasks accordingly. • Adjusts teaching strategies based on GenAI-generated insights from formative assessments.	• Collaborates with GenAI to develop innovative assessment models combining human judgment and machine analytics. • Co-creates multi-dimensional feedback processes that integrate GenAI insights with personalized instruction comments. • Uses GenAI to analyze complex learning data for deeper understanding of student growth and challenges.
Empowering students' usage		
• Shows students how to use GenAI that help them test their own knowledge by answering questions or doing practice exercises. • Encourages students to use GenAI-generated quizzes or reflection questions to see how well they understand the material independently or collaboratively.	• Guides students to interpret GenAI-generated feedback on their work and apply it to revise and improve assignments. • Supports students in using GenAI tools to monitor their progress and identify areas for development.	• Supports students in designing and implementing GenAI-enhanced peer and self-assessment frameworks. • Facilitates reflective evaluation processes where students collaboratively analyze GenAI feedback alongside human input. • Encourages metacognitive awareness by having students critically assess both GenAI-generated and peer feedback.

Appendix E. Application of the GenAI-responsive competency framework: Research proposal development in the research methods course

Course context: A core course for Master's students where the central task is to develop a robust research proposal.
Teacher's goal: To use GenAI as a collaborative partner in the research design process, fostering critical engagement and scholarly rigor.

Competency Dimension	Basic Tier	Intermediate Tier	Advanced Tier
GenAI Literacy	**Teacher:** Demonstrates prompting for generating research questions (e.g., "Generate questions on 'remote work & team cohesion'"), then critiques output for scope and researchability. **Empowering Students:** Tasks students with using GenAI to define a methodological term (e.g., "interpretive phenomenological analysis") and compare it to a textbook definition.	**Teacher:** Uses complex prompts for lesson design (e.g., "Compare grounded theory vs. case study for 'teacher resilience'"). Refines output with expert nuance. **Empowering Students:** Guides students in iterative dialogue to narrow a research interest into a specific, viable question.	**Teacher:** Uses GenAI as a "debating partner" on epistemological tensions (e.g., positivist vs. constructivist critiques) to prepare for advanced seminars. **Empowering Students:** Students document their "Collaborative Research Design Journey" with GenAI, reflecting on prompting strategies, validation, and ethical boundaries.
Curriculum/ Learning Design	**Teacher:** Uses GenAI to generate foundational resources (e.g., "Annotated literature review examples"). **Empowering Students:** Directs students to use GenAI to explore potential theoretical frameworks for their topic.	**Teacher:** Uses GenAI to create adaptive learning materials (e.g., a personalized guide on choosing statistical tests for a struggling student). **Empowering Students:** Peer groups collaboratively use GenAI to create a checklist of methodological weaknesses for peer-review.	**Teacher:** Co-designs a simulated "grant peer-review" exercise with GenAI providing simulated reviewer comments. **Empowering Students:** Students use GenAI to co-design a personalized project management plan (e.g., a Gantt chart) for their thesis timeline.
Teaching and Learning	**Teacher:** Uses GenAI to live-generate a sample dataset for in-class statistical demonstration. **Empowering Students:** Students use GenAI as a "thought starter" to overcome writer's block (e.g., for structuring an introduction).	**Teacher:** Uses GenAI to role-play as an "ethics board reviewer" to critique a sample application, guiding students in rebuttal. **Empowering Students:** Students use GenAI as a "methodology sounding board" to critique and strengthen their chosen methods.	**Teacher:** Facilitates a "methodology triangulation" activity where groups use GenAI to explore mixed-methods approaches. **Empowering Students:** Students engage in sustained dialogue with GenAI to stress-test their research design (e.g., identify threats to validity).

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Competency Dimension	Basic Tier	Intermediate Tier	Advanced Tier
Assessment	**Teacher:** Uses GenAI to generate low-stakes quiz questions on foundational terminology.	**Teacher:** Uses GenAI for a first-pass analysis of a literature review draft, identifying structural or coverage gaps, before adding conceptual feedback.	**Teacher:** Implements a co-creative grading framework, combining their scholarly assessment with GenAI's analysis of methodological rigor.
	Empowering Students: Students use GenAI tools for self-checking writing clarity and grammar.	**Empowering Students:** Students use GenAI for formative feedback on their methodology section's clarity and justification.	**Empowering Students:** Students run AI-enhanced peer review, using GenAI to generate an initial structured critique which they then deepen with their own insights.

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